

Xiang (Emily) Zuo

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SUMMARY

Extensive industrial machine learning analytic and engineering experience. Strong backgrounds of architect design in machine learning/deep learning enterprise products. Skilled in **natural language processing (NLP)**, **large-scale systems**, and **database**. Technical leadership on managing multiple projects and defining technical direction of emerging AI market.

EDUCATION

Stanford Ignite, Stanford University Graduate School of Business (GSB)	Jan. 2024 - Mar. 2024
Ph.D. in Computer Science, University of South Florida, Florida (USF)	2010 - 2016
M.S. in Computer Science, Huazhong University of Science and Technology (HUST), China	2006 - 2009
B.S. in Computer Science, Huazhong University of Science and Technology (HUST), China	2000 - 2004

WORK EXPERIENCE

Senior Technical Engineering Manager October 2023 - Now
Teradata | Santa Clara, CA

Lead Teradata's global AI & ML enterprise product team, setting technical direction and driving execution across Cloud and On-Prem platforms:

- Enterprise vector store – architecting large-scale semantic search and RAG systems for enterprise customers
- Small LMs– use open-source models to power agent systems for on-prem environments, with fast, secure inference
- Agentic AI – automating data science and BI workflows with skill-based, harnessed agents
- ML functions – scaling parallel in-database analytics across Teradata Vantage
- Partner cross-functionally with product, marketing, and sales leadership to align AI/ML roadmap with go-to-market strategy and customer needs

Staff AI Engineer March 2019 - October 2023
Teradata | Santa Clara, CA

- Lead design and development of TeraAnalyticChat, a customer-facing conversational AI product delivering natural-language Q&A and text-to-SQL generation over in-database analytic functions
- Fine-tuned OpenLLaMA into a Teradata-branded LLM using parameter-efficient fine-tuning (qLoRA, Adapter, PEFT) and retrieval-augmented generation (RAG) over synthetic, public, and employee-generated data, pairing a custom search engine with vector embeddings for grounded, low-latency inference
- Applied prompt engineering techniques — instruction tuning, in-context learning, chain-of-thought reasoning — alongside transfer learning on Hugging Face Transformers to improve model accuracy and generalization across product initiatives
- Built a COVID-19 Q&A system combining BERT-based intent classification, ARIMA time-series forecasting, and Elasticsearch semantic search

Senior Machine Learning Engineer Aug. 2016 - Feb. 2019
Teradata | Santa Clara, CA

- Led end-to-end research, design, and development of statistical and ML methods — XGBoost, RandomForest, TF-IDF, Hashing Encodings, GLMNet, NaiveBayesTextClassifier, Z-test, F-test, Chi-square, ANOVA, KNN, VectorDistance, SMOTE, and etc. — on Teradata's **MPP distributed framework**, expanding the in-database analytics function library
- Serve as the **POC site manager** for enterprise customers including Verizon and AT&T
- Drove customer issue resolution by partnering closely with marketing, account, and professional services teams, ensuring rapid turnaround on complex technical escalations
- Mentored summer interns from Stanford University (2017) and UCLA (2018)

SKILLS

Generative AI & LLM Techniques: **Agentic RAG, LLM Fine-Tuning (LoRA/QLoRA, PEFT), Prompt Engineering, Instruction Tuning, Multi-Agent Systems, Agent Skills & Harness Engineering, Vector Search & Embeddings**
ML/AI Frameworks: PyTorch, Hugging Face Transformers, LangChain, LangGraph, Scikit-learn
Agentic AI & Orchestration: Model Context Protocol (MCP), Tool Use / Function Calling, CrewAI, Claude Agent SDK
Evaluation & Observability: LM-as-Judge, LangSmith, DeepEval, MLFlow
Inference & Optimization: Quantization (GPTQ, AWQ, GGUF), Speculative Decoding, Distillation, Pruning, vLLM, Ray Serve
Vector Databases & Retrieval: Pinecone, pgvector, Milvus, Hybrid Search, Reranking
Programming Languages: Python, Java, C/C++, R, UNIX Shell

HONORS AWARDS & PATENTS

Teradata 2024 Team Recognition Award, Company-wide awards	2024
Teradata 2024 Individual Recognition Award, Company-wide awards	2024
An Open-Source Alternative to ChatGPT, Teradata Hack Day 2nd Prize (2nd among 83)	June 2023
Hyperparameter Tuner for Vantage Analytics, Teradata Hack Day 1st Prize (Top 1 among 60)	June 2021
Patent: "Performing Hypersegmentation Analytics at Scale" , Teradata Patent	Dec 2020